Projector-Mass First-Alias Threshold and an Isospectral Cell Obstruction

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Abstract

The projector-density gauge of RH-335 turns the global noisy parity scalar into signed localized cells. We determine the exact first-alias scale of one such cell. On every bounded-phase natural clock, its parity contribution divided by $H_k = kR^{-2k}$ equals $2C_*\lambda^{\eta_\sigma}(\beta R)^{2k}\pi_\sigma(J)(1+o(1))$. Thus the critical projector-mass exponent is $\kappa_{\mathrm{proj}} = \log(\beta R)/\log\lambda$, distinct from the earlier Duhamel stability exponent. A fixed finite partition necessarily has a maximum normalized parity cell diverging to $+\infty$, but the maximizing cell may move with noise and is not identified with a physical boundary/sibling aggregate; raw and alias terms may still cancel it. We complement the scale theorem with an exact three-state family. A nontrivial interval of smooth similarities preserves strict positivity, row-stochasticity, the spectrum, and every power trace, while the parity projector masses and corrected singleton cells drift by explicit nonzero zero-sum vectors. This is a nonphysical finite algebraic obstruction to recovering cells from global spectral data. It does not prove physical signed Duhamel cancellation or a physical normalized obstruction, and it yields no determinant or Riemann-hypothesis conclusion.

1 Inherited projector gauge and first-alias clock

Let K_σ be the noisy folded operator and let $\lambda_-(\sigma) = -(1-\delta_\sigma)$ be its real simple parity eigenvalue, where the archived law is

$$\delta_\sigma = C_*\sqrt{\sigma} + o(\sqrt{\sigma}), \qquad C_* > 0. \tag{1}$$

Write $E_{-,\sigma}$ for the rank-one Riesz projector itself. RH-335 proves that

$$\pi_\sigma(J) = \mathrm{Tr}(M_J E_{-,\sigma}) \tag{2}$$

is a real finite signed measure with total mass one [3]. Allocating the deterministic parity scalar with this noisy density is a frozen gauge, not canonical physical localization or deterministic/noisy projector transport.

Retain the constants and clock from RH-326 [1]:

$$r_H = \frac{17}{20}, \qquad R = \frac{7}{5}, \qquad \beta = \frac{1}{r_H\sqrt{\lambda}}, \tag{3}$$

and

$$\eta_\sigma = k - \frac{\log(1/\sigma)}{2\log\lambda}. \tag{4}$$

Throughout the moving statements, $k = k_\sigma \in \mathbb{N}$, $k \geq 2$, η_σ remains bounded, and $k \to \infty$ as $\sigma \to 0$.

For a measurable cell J, possibly depending on σ, define

$$\mathcal{G}_{\sigma,k}(J) = r_H^{-2k}\{1 - \lambda_-(\sigma)^{2k}\}\pi_\sigma(J), \qquad H_k = kR^{-2k}. \tag{5}$$

This is only the localized parity-gauge constituent of the RH-335 corrected cell. It omits the localized raw defect and the globally subtracted alias packet.

2 Moving projector-mass threshold

Theorem 2.1 (Exact first-alias projector-mass scale)**.** *On every bounded-phase sequence,*

$$\boxed{\frac{\mathcal{G}_{\sigma,k}(J)}{H_k} = 2C_*\lambda^{\eta_\sigma}(\beta R)^{2k}\pi_\sigma(J)\{1+o(1)\}.} \tag{6}$$

The scalar $o(1)$ *is independent of the chosen cell.*

Proof. Because $k\delta_\sigma \to 0$, the uniform parity expansion of RH-326 gives

$$1-(1-\delta_\sigma)^{2k} = 2kC_*\sqrt{\sigma}\{1+o(1)\}. \tag{7}$$

Even order removes the sign of λ_-, so substitution in (5) and division by H_k yield

$$2C_*\sqrt{\sigma}\left(\frac{R}{r_H}\right)^{2k}\pi_\sigma(J)\{1+o(1)\}.$$

Equation (4) gives $\sqrt{\sigma}\lambda^k = \lambda^{\eta_\sigma}$, while $(\beta R)^{2k} = (R/r_H)^{2k}\lambda^{-k}$. Their product is the required scalar in (6). □

Define

$$\boxed{\kappa_{\mathrm{proj}} = \frac{\log(\beta R)}{\log\lambda} = \frac{\log(28/17)}{\log\lambda} - \frac{1}{2}.} \tag{8}$$

The rational bracket $3/2 < u_c < 25/16$ from the physical observation source implies $\lambda = 2u_c(u_c-1) < 225/128 < (28/17)^2$ [2]. Hence $\beta R > 1$ and $\kappa_{\mathrm{proj}} > 0$. Ordinary substitution of the archived decimal for λ gives

$$\kappa_{\mathrm{proj}} = 0.463406944517002\ldots. \tag{9}$$

This decimal is diagnostic, not an interval certificate. It is not the RH-325 sufficient Duhamel weight ceiling $\gamma_* = 0.3503698834605293\ldots$; in fact their separation is exact. The RH-334 algebraic identities imply $\lambda^3 + 4\lambda^2 - 16 = 0$ [2]. This polynomial is strictly increasing on the positive half-line, and its value at $17/10$ is $473/1000 > 0$, so $\lambda < 17/10$. Moreover,

$$\left(\frac{196}{85}\right)^2 - \left(\frac{17}{10}\right)^3 = \frac{116783}{289000} > 0.$$

Since $\lambda > 1$ and

$$\kappa_{\mathrm{proj}} - \gamma_* = \frac{\log\left((R^2/r_H)/\lambda^{3/2}\right)}{\log\lambda}, \qquad \frac{R^2}{r_H} = \frac{196}{85},$$

it follows that $\kappa_{\mathrm{proj}} > \gamma_*$ exactly. The two exponents also govern different objects.

Corollary 2.2 (Negligibility, exact phase conversion, and critical limit)**.** *On every bounded-phase sequence,*

$$\mathcal{G}_{\sigma,k}(J) = o(H_k) \quad\Longleftrightarrow\quad \pi_\sigma(J) = o((\beta R)^{-2k}). \tag{10}$$

Moreover,

$$\boxed{(\beta R)^{-2k} = \sigma^{\kappa_{\mathrm{proj}}}(\beta R)^{-2\eta_\sigma}.} \tag{11}$$

If

$$(\beta R)^{2k}\pi_\sigma(J) \longrightarrow p, \qquad \eta_\sigma \longrightarrow \eta, \tag{12}$$

then

$$\frac{\mathcal{G}_{\sigma,k}(J)}{H_k} \longrightarrow 2C_*\lambda^{\eta}p. \tag{13}$$

Proof. Bounded phase makes $2C_*\lambda^{\eta_\sigma}\{1+o(1)\}$ bounded above and away from zero. This proves (10) from theorem 2.1. From (4), $k = \log(1/\sigma)/(2\log\lambda) + \eta_\sigma$; inserting this into the left side of (11) proves the exact identity. Finally apply (12) to (6). □

3 A forced large parity cell, but not a physical obstruction

Theorem 3.1 (Fixed-partition maximum). *Let $\mathcal{P} = \{J_1, \dots, J_N\}$ be one fixed finite measurable partition. Then*

$$\max_{1 \le i \le N} \pi_\sigma(J_i) \ge \frac{1}{N}. \tag{14}$$

Consequently,

$$\max_{1 \le i \le N} \frac{\mathcal{G}_{\sigma,k}(J_i)}{H_k} \longrightarrow +\infty. \tag{15}$$

The maximizing index may depend on σ. Along every sequence $\sigma_\ell \to 0$, some fixed index recurs as a maximizer on a subsequence, and its normalized parity cell diverges on that subsequence.

Proof. RH-335 gives $\sum_{i=1}^N \pi_\sigma(J_i) = 1$, so the maximum is at least the arithmetic mean $1/N$, even though individual masses may be negative. The scalar in (6) is positive for small noise. Since bounded phase makes λ^{η_σ} uniformly bounded below and $\beta R > 1$, (14) proves (15). The subsequence statement is the pigeonhole principle on the finite index set. □

Remark 3.2 (Cancellation firewall). The theorem concerns only the parity-gauge constituent. It does not identify a maximizing cell with the physical $\mathcal{B} + \mathcal{S}$ aggregate, and it does not prove a physical nonzero normalized obstruction. The localized raw defect can cancel the parity term inside a corrected cell, other cells can cancel in the partition sum, and the first-alias coefficient further subtracts $\mathcal{A}_{k,2k}$. The maximizing index may also move with noise. Thus physical signed $\Delta_B + \Delta_S$ cancellation remains `NOT_TESTABLE`.

4 A strictly positive isospectral Markov family

We now give a separate finite algebraic witness. Reuse the exact RH-335 matrix and parity projector

$$K = \begin{pmatrix} 3/17 & 7/51 & 35/51 \\ 4/85 & 83/255 & 32/51 \\ 58/85 & 1/51 & 76/255 \end{pmatrix}, \tag{16}$$

$$E_- = \begin{pmatrix} 10/17 & -5/51 & -25/51 \\ 8/17 & -4/51 & -20/51 \\ -10/17 & 5/51 & 25/51 \end{pmatrix}. \tag{17}$$

For $t \neq 1$, put

$$S_t = \begin{pmatrix} 1-t & t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad K_t = S_t^{-1} K S_t. \tag{18}$$

Define $q(t) = 35 - 38t - 12t^2$. Exact multiplication gives the audited formula

$$K_t = \begin{pmatrix} \dfrac{15-4t}{85} & \dfrac{q(t)}{255(1-t)} & \dfrac{35-32t}{51(1-t)} \\ \dfrac{4(1-t)}{85} & \dfrac{83+12t}{255} & \dfrac{32}{51} \\ \dfrac{58(1-t)}{85} & \dfrac{5+174t}{255} & \dfrac{76}{255} \end{pmatrix}. \tag{19}$$

Theorem 4.1 (Positive row-stochastic isospectral interval). *For every*

$$\boxed{-\frac{5}{174} < t < \frac{1}{2},} \tag{20}$$

the matrix K_t is strictly positive and row-stochastic. For every such t,

$$\operatorname{spec}(K_t) = \{1, -2/5, 1/5\}, \tag{21}$$

and for every integer $m \geq 1$,

$$\boxed{\operatorname{Tr} K_t^m = 1 + (-2/5)^m + (1/5)^m.} \tag{22}$$

Proof. On (20), every denominator in (19) is positive. The nonconstant numerators

$$15 - 4t, \quad q(t), \quad 35 - 32t, \quad 83 + 12t, \quad 5 + 174t$$

are also positive; for q, it suffices that $t < 1/2$ because its positive root exceeds $1/2$. More directly, $q'(t) = -38 - 24t < 0$ on the stated interval and $q(1/2) = 13$, so $q(t) > 13$. Thus every entry is positive. Since $S_t\mathbf{1} = \mathbf{1}$, also $S_t^{-1}\mathbf{1} = \mathbf{1}$, and $K_t\mathbf{1} = S_t^{-1}K\mathbf{1} = \mathbf{1}$.

Similarity preserves the spectrum and gives $K_t^m = S_t^{-1}K^m S_t$. RH-335 supplies the three distinct eigenvalues of K, so trace invariance proves (22). □

Remark 4.2 (Sufficient versus maximal). The interval (20) is convenient, not maximal. Intersecting all strict entry inequalities in the connected component containing zero gives

$$\left(-\frac{5}{174}, \frac{-19 + \sqrt{781}}{12}\right). \tag{23}$$

The upper endpoint is the positive root of $q(t)$. No maximality claim is made beyond this explicit finite family.

5 Moving projector masses and corrected singleton cells

Let

$$E_-(t) = S_t^{-1}E_-S_t. \tag{24}$$

It is the Riesz projector of K_t at $-2/5$ by similarity.

Proposition 5.1 (Exact projector-mass drift)**.** *For the three singleton windows,*

$$\boxed{\pi(t) = \operatorname{diag} E_-(t) = \left(\frac{10 - 8t}{17}, \frac{-4 + 24t}{51}, \frac{25}{51}\right).} \tag{25}$$

Therefore

$$\pi(t) - \pi(0) = \frac{8t}{17}(-1, 1, 0), \qquad \sum_i \pi_i(t) = 1. \tag{26}$$

Proof. Insert (17) and (18) into (24) and read the diagonal. Subtraction gives (26); its entries sum to zero. □

Use now the RH-335 fixed-order fixture $n = 2$, $r_H = 17/20$, parity eigenvalue $-2/5$, and zero deterministic singleton slots. Define

$$\mathcal{C}_i(t) = \left(\frac{20}{17}\right)^2 \left\{(K_t^2)_{ii} + \frac{21}{25}\pi_i(t)\right\}. \tag{27}$$

Theorem 5.2 (Isospectral corrected-cell obstruction)**.** *The corrected singleton cells are*

$$\boxed{\mathcal{C}(t) = \left(\frac{6800 - 5760t}{4913}, \frac{400 + 5760t}{4913}, \frac{6672}{4913}\right).} \tag{28}$$

For every admissible t,

$$\sum_i \mathcal{C}_i(t) = \frac{48}{17}, \tag{29}$$

$$\mathcal{C}(t) - \mathcal{C}(0) = \frac{5760t}{4913}(-1, 1, 0). \tag{30}$$

At $t = 1/100$, the drift is exactly

$$\left(-\frac{288}{24565}, \frac{288}{24565}, 0\right). \tag{31}$$

Proof. Similarity applied to K^2 gives

$$\operatorname{diag} K_t^2 = \left(\frac{215 - 192t}{425}, \frac{53 + 192t}{425}, \frac{242}{425}\right).$$

Combine this with (25) in (27) and simplify to (28). Summation and subtraction prove (29)–(30); setting $t = 1/100$ gives (31). □

The full spectrum and all power traces are constant, yet the corrected singleton cells move. This proves nonidentification of those cells from global spectral data in a strictly positive row-stochastic family. RH-210 already proved the general logical phenomenon that projectors may move under a fixed divisor [4]. The narrow addition here is the simultaneous positivity, Markov normalization, all-power trace lock, and explicit corrected-cell ledger.

The family remains nonphysical finite algebra. In particular, fixed $n = 2$ is not a $k = 1$ first-alias counterloop: the archived counterloop definition requires $k \geq 2$.

6 Reproduction and claim boundary

The artifact evaluates every matrix identity with exact rational arithmetic. It checks the displayed formulas, the exact rational exponent-separation certificate, strict positivity at rational sample points inside the proved interval, the endpoint factorization, two-sided projector intertwining, twelve power traces, projector masses, corrected cells, invariant sums, and the exact one-percent drift. Floating-point rows only reproduce the symbolic phase conversion and the diagnostic exponent; they are not interval certificates.

RH-336 proves the moving projector-mass threshold and a nonphysical isospectral corrected-cell obstruction. It does not identify a physical $\Delta_B + \Delta_S$ cancellation, prove a physical nonzero normalized obstruction, exclude raw/local/alias signed cancellation, close the far or off-alias terms, transport the noisy head, or glue a determinant. No result constructs a Hilbert–Polya operator, identifies Riemann zeros, proves a von Mangoldt trace formula or completed-zeta divisor equality, or proves the Riemann hypothesis. Gates A–E remain false/open.

References

[1] Bin Wang. Parity-renormalized first-alias packet identity, 2026. Repository paper RH-326.

[2] Bin Wang. Gauge-fixed physical first-alias observation map, 2026. Repository paper RH-334.

[3] Bin Wang. Projector-localized parity ledgers and cellwise extension nonuniqueness, 2026. Repository paper RH-335.

[4] Liang Wang. A divisor-first pivot inside gate a, 2026. Repository paper RH-210.